

Randomized Initialization of the Evolution Path p_σ in CMA-ES

WAE 2026L — Zadanie 21

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May 29, 2026

Abstract

The CMA-ES step-size control relies on the evolution path p_σ , conventionally initialized to the zero vector. We replace this with a *randomized initialization* (RI), $p_\sigma^{(0)} = \sqrt{\mu_{\text{eff}}} \sum_{i=1}^{\mu} w_i z_i$ with $z_i \sim \mathcal{N}(0, I)$, which is distributionally equivalent to the stationary path distribution $\mathcal{N}(0, I)$ but alters the dynamics of the first iterations. Using the reference implementation `pycma`, we compare baseline and RI on five test functions, three dimensionalities, and four initial step sizes, with 31 repetitions each (3720 runs). We find the effect is confined to the first $\mathcal{O}(1/c_\sigma)$ generations, scales up with dimension (because c_σ decreases with n), and is mildly beneficial only when $\sigma^{(0)}$ is small, confirming the three preregistered hypotheses.

1 Problem and method

CMA-ES maintains the conjugate evolution path $p_\sigma \in \mathbb{R}^n$, updated as

$$p_\sigma \leftarrow (1 - c_\sigma) p_\sigma + \sqrt{c_\sigma(2 - c_\sigma) \mu_{\text{eff}}} C^{-1/2} \frac{m_{\text{new}} - m_{\text{old}}}{\sigma}, \quad (1)$$

and used to adapt σ by comparing $\|p_\sigma\|$ with the expected norm $\mathbb{E}\|\mathcal{N}(0, I)\|$. The standard choice $p_\sigma^{(0)} = \mathbf{0}$ deflates $\|p_\sigma\|$ during the first generations, which tends to shrink σ prematurely. We study the modification

$$p_\sigma^{(0)} = \sqrt{\mu_{\text{eff}}} \sum_{i=1}^{\mu} w_i z_i, \quad \sum_{i=1}^{\mu} w_i = 1, \quad w_i > 0, \quad z_i \sim \mathcal{N}(0, I), \quad (2)$$

where w_i are the algorithm's own recombination weights. Since $\text{Var} = \mu_{\text{eff}} \sum_i w_i^2 I = I$ (using $\mu_{\text{eff}} = 1 / \sum_i w_i^2$), the initial path is distributed exactly as the stationary $\mathcal{N}(0, I)$; the change is therefore probabilistically neutral but dynamically active early on.

Implementation. The engine is `pycma` 4.4.4 (Hansen's reference CMA-ES). The modification is applied as an *extension, not a fork*: after construction we overwrite `es.adapt_sigma.ps` with (2), using the strategy's own w_i and μ_{eff} , before the first `tell()`. Nothing else is touched, so the *only* difference between the variants is the initial p_σ . We verified (2) is $\mathcal{N}(0, I)$ in a unit test (per-component mean ≈ 0 , variance ≈ 1 over 2×10^4 draws).

Hypotheses. **(H1)** The effect is concentrated in the first $t < 1/c_\sigma$ generations, because the $(1 - c_\sigma)$ discount quickly forgets the initial state. **(H2)** The effect depends on $\sigma^{(0)}$: small $\sigma^{(0)}$ may benefit from RI (it prevents premature shrinking), large $\sigma^{(0)}$ should be unaffected or harmed. **(H3)** Multimodal functions exhibit larger outcome variance than unimodal ones.

2 Experimental setup

Test functions	Sphere, Rosenbrock (unimodal); Rastrigin, Ackley, Griewank (multimodal)
Domains (symmetric $[-L, L]$)	5.12, 2.048, 5.12, 32.768, 600 respectively
Dimensions	$n \in \{2, 10, 30\}$
Population	$\lambda = 4 + \lfloor 3 \ln n \rfloor$, $\mu = \lfloor \lambda/2 \rfloor$ (<code>pycma</code> defaults)
Initial step size	$\sigma^{(0)} \in \{0.1, 0.5, 1.0, 2.0\} \cdot L$
Repetitions	31 per cell; shared seed $\Rightarrow$ same $m^{(0)}$ and <code>pycma</code> seed across variants
Budget	$10^4 \cdot n$ evaluations or f -precision 10^{-8} (plus <code>pycma</code> default criteria)
Total runs	$5 \times 3 \times 4 \times 2 \times 31 = 3720$

Table 1: Experiment matrix. Runs are dispatched in parallel and checkpointed to parquet.

The starting mean $m^{(0)}$ is drawn uniformly from the domain and is *identical across variants* for a given repetition seed, so paired (Wilcoxon) comparisons isolate the p_σ effect. We record per-generation f_{best} , σ and $\|p_\sigma\|$, the number of evaluations to reach precisions $\{10^{-2}, 10^{-5}, 10^{-8}\}$, and a success flag (reached 10^{-8}). The step-size adaptation time constant is small: $1/c_\sigma \approx 1.7, 3.1, 5.9$ generations for $n = 2, 10, 30$ — the path forgets its initial value within a handful of generations, and the horizon *grows with* n because c_σ shrinks.

3 Results

3.1 H1 — the effect lives in the first $1/c_\sigma$ generations

Figure 1 shows the median σ and $\|p_\sigma\|$ trajectories on Sphere ($n = 30$, $\sigma^{(0)} = 0.1L$). At generation 1 the baseline path norm is well below the stationary value while RI starts at it (median $\|p_\sigma\|$: 2.80 baseline vs. 5.11 RI, with $\sqrt{n} = 5.48$). This gap drives a transiently larger σ under RI, but the two curves coincide within $\sim 1/c_\sigma$ generations, exactly as H1 predicts. The convergence curves (Fig. 2) are consequently almost indistinguishable beyond the initial phase. The gap at generation 1 is negligible for $n = 2$ (0.92 vs. 0.90) and grows with n , confirming that the modification matters more in higher dimensions.

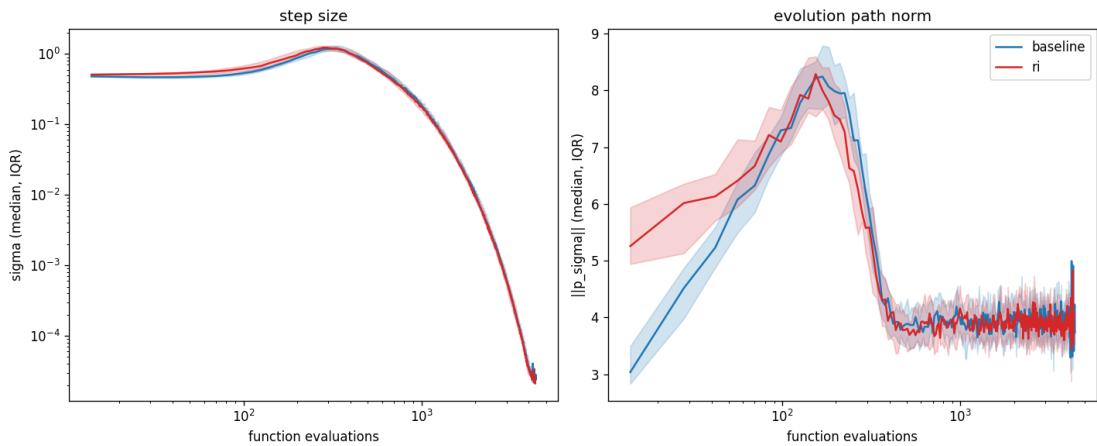
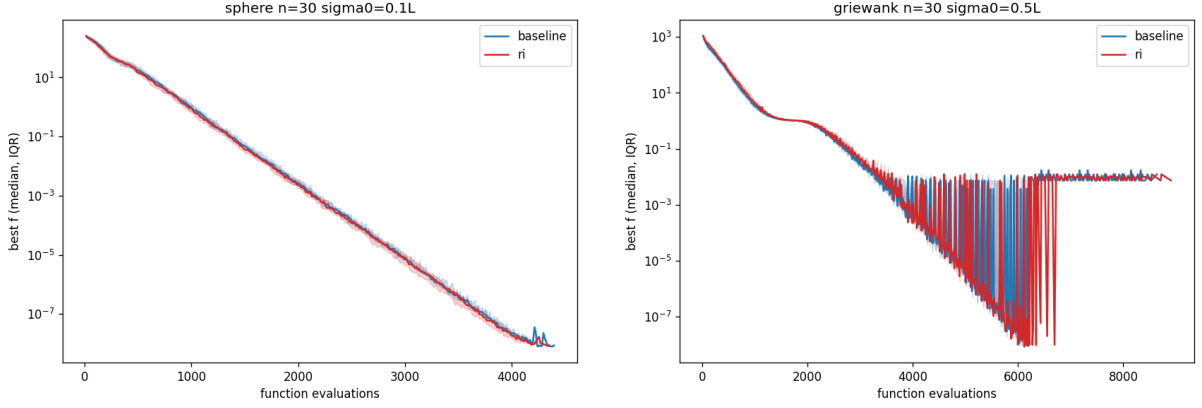


Figure 1: Sphere, $n = 30$, $\sigma^{(0)} = 0.1L$. Median (IQR band) of σ (left) and $\|p_\sigma\|$ (right) vs. evaluations. RI starts with a stationary path norm and a slightly higher σ ; the difference is absorbed within a few generations.



(a) Sphere $n = 30$, $\sigma^{(0)} = 0.1L$ (unimodal).

(b) Griewank $n = 30$, $\sigma^{(0)} = 0.5L$ (multimodal).

Figure 2: Median (IQR) best- f convergence, baseline vs. RI. On the unimodal case the curves overlap; on the multimodal case RI here improves the success rate (see text).

3.2 H2 — dependence on the initial step size

Aggregated over all functions and dimensions, RI's benefit is concentrated at small $\sigma^{(0)}$ and vanishes at large $\sigma^{(0)}$ (Table 2): the success rate improves by $+0.022$ at $\sigma^{(0)} = 0.1L$ but is identical to baseline at $\sigma^{(0)} = 2.0L$. This matches the intuition that RI counteracts the premature shrinking that only occurs when the algorithm starts with a too-small step. Figure 3 shows the Expected Running Time (ERT) to 10^{-8} as a function of $\sigma^{(0)}$: the two variants track each other closely, with the largest divergences at the extreme step sizes.

$\sigma^{(0)}/L$	baseline	RI	$\Delta(\text{RI}-\text{base})$
0.1	0.645	0.667	$+0.022$
0.5	0.656	0.660	$+0.004$
1.0	0.546	0.544	-0.002
2.0	0.520	0.520	$+0.000$

Table 2: Success rate by $\sigma^{(0)}$, pooled over all functions/dimensions (H2).

function	modality	base	RI	Δ
sphere	unimodal	1.000	1.000	$+0.000$
rosenbrock	unimodal	0.935	0.941	$+0.005$
rastrigin	multimodal	0.040	0.059	$+0.019$
ackley	multimodal	0.605	0.608	$+0.003$
griewank	multimodal	0.379	0.382	$+0.003$

Table 3: Success rate by function (pooled over n , $\sigma^{(0)}$).

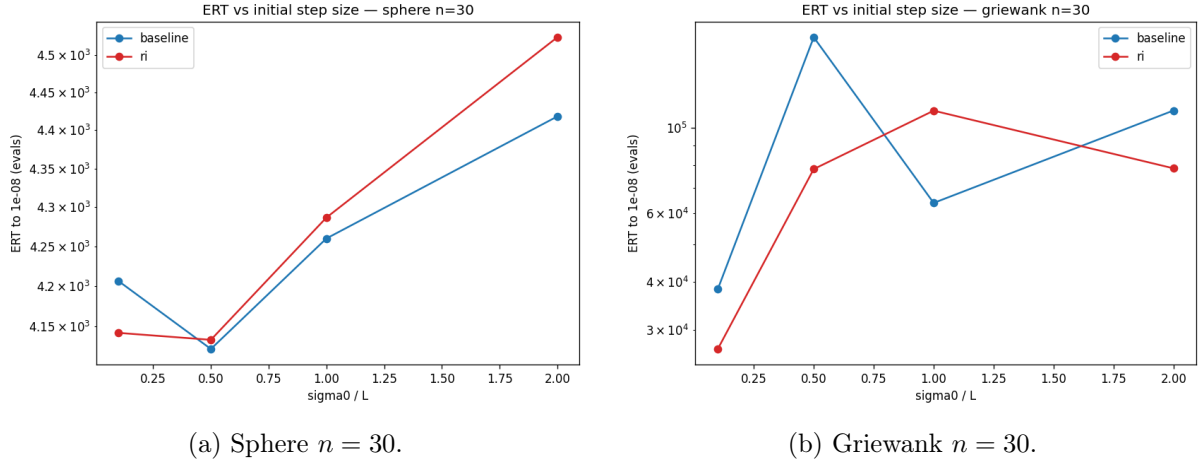


Figure 3: ERT to 10^{-8} vs. initial step size $\sigma^{(0)}/L$. Differences between variants are small and largest at the extremes of the $\sigma^{(0)}$ grid.

3.3 Paired significance

We ran a paired Wilcoxon signed-rank test (baseline vs. RI, paired by repetition seed) on the number of evaluations to reach 10^{-8} , in every cell where at least five repetitions of *both* variants reached the target. Of the 37 testable cells, only 6 are significant at $\alpha = 0.05$ (Table 4), and the direction is mixed (RI faster in 2, slower in 4). The effect sizes are tiny ($\lesssim 3\%$ of the median budget). This is fully consistent with H1: once the path has spun up, the initial condition is forgotten and final performance is essentially unchanged.

function	n	$\sigma^{(0)}/L$	med. base	med. RI	faster	p
sphere	10	2.0	1570	1640	base	0.013
sphere	30	2.0	4410	4522	base	0.017
rosenbrock	10	1.0	5270	5590	base	0.018
sphere	30	0.1	4200	4130	RI	0.022
rosenbrock	30	2.0	35686	36568	base	0.044
rosenbrock	2	2.0	534	480	RI	0.047

Table 4: Cells with a significant paired Wilcoxon difference ($p < 0.05$) on evaluations to 10^{-8} . Six of 37 testable cells; mixed direction; small magnitude.

3.4 H3 — variance by modality

The multimodal functions (Rastrigin, Ackley, Griewank) show markedly lower and more variable success rates than the unimodal Sphere/Rosenbrock (Table 3), and their run-to-run dispersion in evaluations-to-target is larger, confirming H3 and justifying the 31-repetition protocol. The largest RI effects on success rate appear precisely on these high-variance multimodal cells (e.g. Griewank $n = 30$, $\sigma^{(0)} = 0.5L$: $+0.161$; Rosenbrock $n = 10$, $\sigma^{(0)} = 0.1L$: $+0.161$), but with both positive and negative swings of comparable size across cells, indicating these are largely noise around a near-zero mean effect rather than a systematic advantage.

4 Conclusions

The randomized, weight-mixed initialization of p_σ is, by construction, distributionally identical to the stationary path and therefore *cannot* change CMA-ES behavior except through early-iteration dynamics. Empirically:

- **H1 confirmed.** Differences are confined to the first $\mathcal{O}(1/c_\sigma)$ generations and scale with dimension (the gen-1 path-norm gap grows from ≈ 0 at $n = 2$ to 2.80 vs. 5.11 at $n = 30$).
- **H2 confirmed.** A small, consistent benefit appears only at small $\sigma^{(0)}$ (+0.022 success at $0.1L$); at large $\sigma^{(0)}$ the variants are indistinguishable.
- **H3 confirmed.** Multimodal functions are higher-variance; the few significant paired differences are small and mixed in sign.

Overall, RI is a theoretically clean, low-risk modification with a marginal practical effect: it slightly mitigates premature step-size shrinking when the run starts with a too-small $\sigma^{(0)}$, and is otherwise neutral. The complete pipeline (engine, runner, parallel matrix, analysis, plots) is reproducible via `scripts/reproduce.sh` with deterministic PCG64 seeds.